

Skanda Bhairava

CV

About me

Computer Science student interested in systems, and simulations. I enjoy learning by building and I am looking for internships where I can enhance my skill set while contributing to the organization's goals.

Contact

@ skandabhairava@gmail.com
skandabhairava@gmail.com

skandabhairava
github.com/skandabhairava

skandabhairava
linkedin.com/in/skandabhairava

+91 (948) 066-5198

Bangalore, India

Languages

Python • Go • Cpp • Rust • JS/TS

Tools

PyTorch • PyGame • RayCore
• Svelte • FastAPI • Gin • P5JS

Interests

Distributed Systems • Simulations • Graphics • Research

EXPERIENCE

June 2025 **Kaynes Technology**

SOFTWARE ENGINEER INTERN • Mysuru, India

Proactively designed and shipped two internal tools for RFID configuration and UI debugging, both of which were adopted by the RnD team. Efforts were later formally recognized as a short internship.

EDUCATION

2023–Current **Bachelors •**

INFORMATION TECHNOLOGY (HONOURS)

Manipal Institute of Technology Bangalore

CGPA: 9.29/10

HONOURS AND AWARDS

2026 **Best Research Poster Award, Turinger Symposium**

MANIPAL INSTITUTE OF TECHNOLOGY • Bangalore

Received the Best Poster Award for research, at the Turinger Symposium, Manipal Institute of Technology Bangalore, February 2026.

2026 **Hackathon Finalist**

HCL GUVI • AI Impact Summit 2026, Delhi

Our team was placed in the top 2% in the HCL GUVI hackathon for developing a Honeypot to extract details from a Scammer in a textual environment.

2024, 2025 **Manipal Achievers Scholarship (Dean's List)**

MANIPAL ACADEMY OF HIGHER EDUCATION • Bangalore

Received the Manipal Achievers Scholarship in 2024 and 2025 for maintaining a CGPA of 9.29 out of 10.

PUBLICATIONS

Skanda M Bhairava, Gururaj H L. "ADaRES: Adaptive Data Reallocation and Epoch Scheduling in Federated Learning Under Network and Resource Constraints". COMSNETS 2026
doi.org/10.1109/COMSNETS67989.2026.11418093

PROJECTS (TOP 6)

HierarchicalFLSim

Python • RayCore • Systems

Built a configurable Hierarchical Federated Learning simulator used to experiment with flexible topology, Non-IID distributions, client dropout and custom aggregation algorithms. The simulator also helps validate Cost model for hierarchical training.

Leaf Disease Detection

Python • Pytorch • ML

Developed a lightweight CNN that performed better than baseline architectures (AlexNet, DeepNet) on a leaf disease dataset in important metrics.

Raytracer

C++ • Multithreading • Graphics

Implemented a 3D Ray Tracing Engine from scratch in C++ using hand-derived math, multithreading and a custom Pixel Representation Format.

GravBall

Python • PyGame • Simulation

Designed and Optimized a Physics based Python game to run at 120 fps by redesigning collision and rendering loops.

Obbattu

Rust, TS • Svelte, OracleVPS • Full Stack

Built a full stack crossword generator with daily puzzle regeneration using a backtracking solver, with an in-built recovery mechanism after host downtimes. Deployed on VPS with Rust backend and Svelte frontend.

DragonCommissions

Python • Automation • Management

Co-founded a plugin development hub that built 100+ custom plugins for gaming creators. Coordinated a team of around 20 developers, and handled client requirements for around 10+ creators. I also built internal automation tools and pipelines that made development workflows efficient.